

Computer Graphics — Lecture 2

Math Review Part I: Linear Algebra — Study Questions

Batch 1 - Questions 1 to 20: Why Linear Algebra, and What a Vector Is

Q1. Why is linear algebra described as important for computer graphics?

- a) It is only used for drawing text
- b) It is only needed for hardware design
- c) It replaces the need for any programming
- d) It is an effective bridge between geometry, physics, etc., and computation

Answer: d) It is an effective bridge between geometry, physics, etc., and computation

Q2. According to the lecture, once you can express a graphics problem in terms of linear algebra, you are essentially done because:

- a) The GPU automatically draws the result
- b) The problem no longer needs any input data
- c) The answer is always a single number
- d) You can ask the computer to solve $Ax = b$

Answer: d) You can ask the computer to solve $Ax = b$

Q3. Which of the following is listed as an area made possible by fast numerical linear algebra?

- a) Image processing, physically-based animation, and geometry processing
- b) Only network routing
- c) Only database indexing
- d) Only file compression

Answer: a) Image processing, physically-based animation, and geometry processing

Q4. Linear algebra is defined in the lecture as the study of:

- a) Only 3D rotations
- b) Vector spaces and linear maps between them
- c) Polynomials and nothing else
- d) Only matrices of integers

Answer: b) Vector spaces and linear maps between them

Q5. What is the intuitive mental model of a vector used in the lecture?

- a) A closed curve
- b) A grid of pixels
- c) A little arrow
- d) A single point

Answer: c) A little arrow

Q6. Which of the following is mentioned as data in graphics that may not look like arrows but still behaves like vectors?

- a) Only file names
- b) Only integers
- c) Polynomials, images, and radiance

d) Only text strings

Answer: c) Polynomials, images, and radiance

Q7. What information does a vector fundamentally encode?

- a) Color and texture
- b) Only its starting point
- c) Position and mass
- d) Direction and magnitude

Answer: d) Direction and magnitude

Q8. A vector in 2D can be encoded by a length and an angle relative to a fixed direction. This is called:

- a) Barycentric coordinates
- b) Homogeneous coordinates
- c) Polar coordinates
- d) Cartesian coordinates

Answer: c) Polar coordinates

Q9. Traditionally, a vector does NOT include a:

- a) Basepoint
- b) Direction
- c) Length
- d) Magnitude

Answer: a) Basepoint

Q10. A vector with a basepoint is sometimes called a:

- a) Tangent vector
- b) Normal vector
- c) Unit vector
- d) Zero vector

Answer: a) Tangent vector

Q11. Measuring the components of a vector with respect to a chosen coordinate system gives which representation (named after Descartes)?

- a) Spherical coordinates only
- b) Cartesian coordinates
- c) Polar coordinates
- d) Radiance coordinates

Answer: b) Cartesian coordinates

Q12. What does the lecture warn about coordinates?

- a) Only integer coordinates are valid
- b) You cannot directly compare coordinates in different systems, e.g., (r, θ) with (x, y)
- c) Coordinates can never be used on a computer
- d) Coordinates are always identical in every system

Answer: b) You cannot directly compare coordinates in different systems, e.g., (r, θ) with (x, y)

Q13. The first basic operation on vectors shown in the lecture is addition. How is it done geometrically?

- a) Rotating one arrow by 90 degrees

- b) Taking the longer of the two
- c) Multiplying their lengths
- d) Placing the arrows end to end

Answer: d) Placing the arrows end to end

Q14. The fact that $u + v = v + u$ means vector addition is:

- a) Non-linear
- b) Associative only
- c) Commutative (abelian)
- d) Distributive

Answer: c) Commutative (abelian)

Q15. The term 'abelian' used for commutative addition comes from the name of:

- a) René Descartes
- b) Isaac Newton
- c) Niels Henrik Abel
- d) Carl Friedrich Gauss

Answer: c) Niels Henrik Abel

Q16. The second basic operation on vectors is:

- a) Sorting the components
- b) Dividing one vector by another
- c) Scaling by a number (scalar)
- d) Squaring a vector

Answer: c) Scaling by a number (scalar)

Q17. Which identity describes how scaling behaves with repeated scaling?

- a) $a(bu) = (a + b)u$
- b) $a(bu) = abu^2$
- c) $a(bu) = a + bu$
- d) $a(bu) = (ab)u$

Answer: d) $a(bu) = (ab)u$

Q18. Which identity describes the interaction of addition and scaling?

- a) $a(u + v) = a + u + v$
- b) $a(u + v) = au \cdot av$
- c) $a(u + v) = au + av$
- d) $a(u + v) = a(uv)$

Answer: c) $a(u + v) = au + av$

Q19. According to the lecture, where do the rules (axioms) of a vector space come from?

- a) They were given by an authority with no explanation
- b) They come only from computer hardware
- c) They are random conventions
- d) The geometric behavior of little arrows

Answer: d) The geometric behavior of little arrows

Q20. Any collection of objects satisfying all of the vector-space properties is called:

- a) An invalid set
- b) A matrix
- c) A vector space, even if the objects do not look like little arrows

d) A scalar field only

Answer: c) A vector space, even if the objects do not look like little arrows

Batch 2 - Questions 21 to 38: Euclidean Space, Functions as Vectors, and Coordinates

Q21. The most common example of a vector space, denoted $\mathbb{R}^n$, means:

- a) n functions
- b) n complex numbers
- c) n integers only
- d) n real numbers

Answer: d) n real numbers

Q22. The tuple $(1.23, 4.56, \pi/2)$ is a point in which space?

- a) $\mathbb{R}^3$
- b) $\mathbb{R}^1$
- c) $\mathbb{R}^4$
- d) $\mathbb{R}^2$

Answer: a) $\mathbb{R}^3$

Q23. Why is Euclidean n -dimensional space such a common example? (choose the best answer)

- a) It is the only vector space that exists
- b) It cannot represent images
- c) It looks a lot like the space we live in and is easy to encode on a computer as a list of floating-point numbers
- d) It needs no memory to store

Answer: c) It looks a lot like the space we live in and is easy to encode on a computer as a list of floating-point numbers

Q24. Which of the following is given as another very important example of vector spaces in graphics?

- a) Spaces of file names
- b) Spaces of passwords
- c) Spaces of functions
- d) Spaces of network packets

Answer: c) Spaces of functions

Q25. Why are spaces of functions important in computer graphics?

- a) They replace the need for coordinates
- b) Many objects we work with in graphics are functions
- c) Functions are always linear
- d) Functions never change

Answer: b) Many objects we work with in graphics are functions

Q26. Which of the following is listed as an example of a function that behaves as a vector?

- a) Only the size of a file
- b) Only memory addresses
- c) Only the keyboard state
- d) Images, radiance from a light source, surfaces, and modal vibrations

Answer: d) Images, radiance from a light source, surfaces, and modal vibrations

Q27. How can two functions be added, according to the lecture?

- a) Add their values at each point x
- b) Multiply their graphs
- c) Take the maximum of the two graphs
- d) It is not possible to add functions

Answer: a) Add their values at each point x

Q28. What is the 'zero vector' in a space of functions?

- a) The function equal to one for all x
- b) There is no zero vector
- c) The function equal to zero for all x
- d) The function x

Answer: c) The function equal to zero for all x

Q29. Short answer given in the lecture: are functions vectors?

- a) No, only arrows are vectors
- b) Only functions on $[0,1]$ are
- c) Yes, even if they do not look like little arrows
- d) Only polynomial functions are

Answer: c) Yes, even if they do not look like little arrows

Q30. After coming up with a rule for adding pairs of numbers, how does the lecture check that it faithfully encodes the geometry of little arrows?

- a) Test it on a single example only
- b) Check that it agrees with the list of rules the arrows must obey
- c) Draw it once and trust it
- d) Assume it works since it looks simple

Answer: b) Check that it agrees with the list of rules the arrows must obey

Q31. How are vectors added in Cartesian coordinates?

- a) Component by component: $(u_1, u_2) + (v_1, v_2) = (u_1 + v_1, u_2 + v_2)$
- b) By adding the lengths only
- c) By multiplying the components
- d) By adding only the first components

Answer: a) Component by component: $(u_1, u_2) + (v_1, v_2) = (u_1 + v_1, u_2 + v_2)$

Q32. According to the lecture, why is turning geometric observations into algebraic rules convenient?

- a) It makes vectors non-commutative
- b) It is convenient for symbolic manipulation and numerical computation
- c) It is required by all graphics hardware
- d) It removes the need for geometry forever

Answer: b) It is convenient for symbolic manipulation and numerical computation

Q33. What should you always ask about a rule given to you by an authority?

- a) Is it short enough to memorize?
- b) Who published it first?
- c) Where does this rule come from, and what does it mean geometrically (can you draw a picture)?
- d) Is it written in the textbook?

Answer: c) Where does this rule come from, and what does it mean geometrically (can you draw a picture)?

Q34. How do we scale a vector in coordinates?

- a) Divide the scalar by each component
- b) Add the scalar to each component
- c) Multiply each component by the scalar
- d) Multiply only the first component

Answer: c) Multiply each component by the scalar

Q35. What is $(3/2) \cdot (4, 2)$?

- a) (6, 3)
- b) (4, 3)
- c) (5.5, 3.5)
- d) (12, 6)

Answer: a) (6, 3)

Q36. The midpoint m of two points a and b is computed as:

- a) $m = 2(a + b)$
- b) $m = a - b$
- c) $m = \frac{1}{2}(a + b)$
- d) $m = a \cdot b$

Answer: c) $m = \frac{1}{2}(a + b)$

Q37. What is the midpoint of $a = (3, 4)$ and $b = (7, 2)$?

- a) (10, 6)
- b) (5, 3)
- c) (4, 2)
- d) (5, 6)

Answer: b) (5, 3)

Q38. The midpoint example is used to show that:

- a) Vectors cannot be added to points
- b) Combining vector operations builds up operations needed for computer graphics
- c) Midpoints cannot be computed with vectors
- d) Scaling is not a valid operation

Answer: b) Combining vector operations builds up operations needed for computer graphics

Batch 3 - Questions 39 to 59: Measuring Vectors and Norms

Q39. What two quantities does a vector encode that we want to measure?

- a) Orientation (direction) and magnitude
- b) Origin and destination only
- c) Mass and velocity
- d) Color and brightness

Answer: a) Orientation (direction) and magnitude

Q40. The number $|v|$ assigned to a vector v is called its:

- a) Basepoint

- b) Scalar field
- c) Component
- d) Length, magnitude, or norm

Answer: d) Length, magnitude, or norm

Q41. Intuitively, the norm of a vector should capture:

- a) Where it starts
- b) How 'big' the vector is
- c) Which direction it points
- d) How many components it has

Answer: b) How 'big' the vector is

Q42. Which natural property says the norm should not be negative?

- a) Commutativity
- b) Positivity: $|u| \geq 0$
- c) Triangle inequality
- d) Homogeneity: $|cu| = |c||u|$

Answer: b) Positivity: $|u| \geq 0$

Q43. A norm should be zero only for:

- a) Every unit vector
- b) The zero vector
- c) Any vector in $\mathbb{R}^2$
- d) Every vector of length one

Answer: b) The zero vector

Q44. If a vector is scaled by a factor c , its norm should:

- a) Stay the same
- b) Become c^2
- c) Scale by the same amount: $|cu| = |c||u|$
- d) Become negative

Answer: c) Scale by the same amount: $|cu| = |c||u|$

Q45. Which norm property expresses that the shortest path between two points is a straight line?

- a) $|u + v| \leq |u| + |v|$
- b) $|u + v| = 0$
- c) $|u + v| \geq |u| + |v|$
- d) $|u + v| = |u| \cdot |v|$

Answer: a) $|u + v| \leq |u| + |v|$

Q46. The slide says the final norm property is sometimes called the 'pentagon inequality' because:

- a) It only applies in five dimensions
- b) The diagram looks like a pentagon
- c) It involves five vectors
- d) It was discovered by a mathematician named Penta

Answer: b) The diagram looks like a pentagon

Q47. According to the formal definition, a norm is:

- a) Only the length of an arrow in 2D

- b) Any function of a vector, with no conditions
- c) Any function that assigns a number to each vector and satisfies the norm properties for all vectors u, v and all scalars a
- d) Only the square root of a sum of squares

Answer: c) Any function that assigns a number to each vector and satisfies the norm properties for all vectors u, v and all scalars a

Q48. Each norm rule has a concrete geometric picture that explains:

- a) Which hardware to use
- b) Why the rule is there
- c) How to draw a texture
- d) How to avoid using it

Answer: b) Why the rule is there

Q49. What is the standard norm of n -vectors called?

- a) The Manhattan norm
- b) The Euclidean norm
- c) The polar norm
- d) The L2 function norm

Answer: b) The Euclidean norm

Q50. The Euclidean norm of $u = (u_1, \dots, u_n)$ is:

- a) The product of the u_i
- b) The square root of the sum of u_i^2 ($i = 1$ to n)
- c) The largest u_i
- d) The sum of the u_i

Answer: b) The square root of the sum of u_i^2 ($i = 1$ to n)

Q51. What is the Euclidean norm of $u = (4, 2)$?

- a) 6
- b) $2\sqrt{5}$
- c) 8
- d) $\sqrt{6}$

Answer: b) $2\sqrt{5}$

Q52. The L^2 norm of functions measures:

- a) The total magnitude of a function
- b) The slope at a single point
- c) The maximum input value
- d) The number of zeros of a function

Answer: a) The total magnitude of a function

Q53. The L^2 norm in the lecture is defined for functions on which domain?

- a) All functions on the real line with no conditions
- b) Only integer-valued functions
- c) Only functions on $[0,10]$
- d) Real-valued functions on the unit interval $[0,1]$ whose square has a well-defined integral

Answer: d) Real-valued functions on the unit interval $[0,1]$ whose square has a well-defined integral

Q54. The L^2 norm of a function f on $[0,1]$ is defined as:

- a) The sum of f at the endpoints
- b) The integral of $f(x)$ from 0 to 1
- c) The maximum of $f(x)$
- d) The square root of the integral of $f(x)^2$ from 0 to 1

Answer: d) The square root of the integral of $f(x)^2$ from 0 to 1

Q55. How does the L^2 norm differ from the Euclidean norm, as described in the lecture?

- a) We replaced the square root with a logarithm
- b) There is no relationship between them
- c) We replaced squares with cubes
- d) We just replaced a sum with an integral

Answer: d) We just replaced a sum with an integral

Q56. For $f(x) = \sqrt{3} \cdot x$ on $[0,1]$, what is $\|f\|^2$?

- a) $\sqrt{3}$
- b) 3
- c) $\int_0^1 3x^2 dx = 1$
- d) 0

Answer: c) $\int_0^1 3x^2 dx = 1$

Q57. What is the L^2 norm $\|f\|$ of $f(x) = \sqrt{3} \cdot x$ on $[0,1]$?

- a) $\sqrt{3}$
- b) 3
- c) 0
- d) 1

Answer: d) 1

Q58. Which notation does the lecture use for the norm of a function versus the norm of a vector in $\mathbb{R}^n$?

- a) Both use $\|\cdot\|$
- b) Both use $|\cdot|$
- c) $|\cdot|$ for a function and $\|\cdot\|$ for a vector
- d) $\|\cdot\|$ for a function and $|\cdot|$ for a vector in $\mathbb{R}^n$

Answer: d) $\|\cdot\|$ for a function and $|\cdot|$ for a vector in $\mathbb{R}^n$

Q59. What does the lecture say about how most integrals in graphics are calculated?

- a) All are calculated by hand exactly
- b) They are never needed in graphics
- c) They are always replaced by matrices
- d) Most are not calculated analytically like this; numerical integration will be discussed later

Answer: d) Most are not calculated analytically like this; numerical integration will be discussed later

Essay Questions

Q60. Why is linear algebra important for computer graphics, according to the lecture?

Answer:

- It is an effective bridge between geometry, physics, etc., and computation.
- In many areas of graphics, once a problem is expressed in linear algebra, you are essentially done: you ask the computer to solve $Ax = b$.

- Fast numerical linear algebra has made modern computer graphics possible (image processing, physically-based animation, geometry processing).

Q61. Explain the two basic vector operations and the rules they obey, and where those rules come from. Why can functions also be treated as vectors?

Answer:

Two basic operations:

- Addition: place vectors end to end. It is commutative (abelian): $u + v = v + u$.
- Scaling: multiply a vector by a scalar a to get au . It behaves as expected, e.g., $a(bu) = (ab)u$.
- Interaction: $a(u + v) = au + av$.

Where the rules come from:

- Each rule comes from the geometric behavior of 'little arrows'; the rules did not 'fall out of the sky'.
- Any collection of objects satisfying all the properties is a vector space, even if the objects do not look like arrows.

Functions as vectors:

- Many objects in graphics are functions (images, radiance from a light source, surfaces, modal vibrations).
- Functions can be added and scaled, and the other properties hold too (e.g., the zero vector is the function equal to zero for all x), so functions are vectors.

Q62. What properties should a norm satisfy? Give the Euclidean norm and the L^2 norm, with one worked example of each.

Answer:

Natural properties of a norm:

- Positivity: $|u| \geq 0$, and $|u| = 0$ only for the zero vector.
- Scaling: $|cu| = |c||u|$.
- Triangle inequality (the 'shortest path is a straight line' property): $|u + v| \leq |u| + |v|$.

Euclidean norm (vectors in $\mathbb{R}^n$):

- $|u| = \sqrt{\sum u_i^2}$. Example: $u = (4, 2)$ gives $|u| = \sqrt{4^2 + 2^2} = 2\sqrt{5}$.

L^2 norm (functions on $[0, 1]$):

- $\|f\| = \sqrt{\int_0^1 f(x)^2 dx}$, i.e., the Euclidean norm with a sum replaced by an integral.
- Example: $f(x) = \sqrt{3} \cdot x$ gives $\|f\|^2 = \int_0^1 3x^2 dx = [x^3]_0^1 = 1$, so $\|f\| = 1$.